E-Commerce Sales Analytics

Power BI Assignment 2

Detailed observations only — no screenshots

Project objective

Build an interactive Power BI report to analyze sales, targets, profit, order counts, monthly trends, and geographic distribution using calculated columns, DAX measures, KPI cards, charts, a matrix, a treemap, a map, and slicers.

Accuracy note

The following observations describe the work represented in the provided screenshot sequence. The visible Sales target table contains Category, Month of Order Date, and Target; a Segment field is not visible. Therefore, the Minimum Target card is documented as the overall minimum target, not minimum target by segment.

1. Review the Order Details table

Observation: Reviewed the available fields in the Order Details table before creating calculations. The fields used in the project include Order ID, Amount, Profit, Quantity, Category, Sub-Category, Profit Margin, and Profit Status. This initial review helped identify the columns needed for sales and profit analysis.

2. Create the Category Type calculated column

Observation: Created a calculated column named Category Type by combining Category and Sub-Category into one descriptive text value. This gives each record a combined category label that can help with grouping and interpreting product-level results in report visuals.

3. Create the Revenue per Order calculated column

Observation: Created Revenue per Order by multiplying Amount by Quantity. The calculated column adds a derived value to each row, allowing the report to use the result in further analysis. The calculation depends on the Amount and Quantity values being valid numeric data.

4. Create the Sales Category calculated column

Observation: Created Sales Category with IF logic to compare each row's Amount with the average Amount. The result labels records as Above Average or Below Average according to the comparison. This classification can be used to group and compare sales records.

5. Create the Order Count measure

Observation: Created an Order Count measure using DISTINCTCOUNT on Order ID. This counts unique order IDs, rather than counting every detail row, which is important when an order can have multiple product lines.

6. Create the Average Profit in Delhi measure

Observation: Created the Average Profit in Delhi measure using CALCULATE and AVERAGE of Profit, with a location filter based on Delhi in the List of Orders table. The measure is intended to return the average profit for Delhi-related records and is displayed in a KPI card.

7. Add a Card for Average Profit in Delhi

Observation: Added a Card visual to the report canvas and assigned the Average Profit in Delhi measure to its value. This converted the measure into a single KPI number that can be read quickly without scanning a table or chart.

8. Design the report header and page background

Observation: Styled the report page with a dark-blue background and a blue header area. Decorative images and the heading DAX CALCULATIONS were placed in the header to create a consistent visual identity for the report page.

Data preparation and DAX

9. Create the YTD Sales measure

Observation: Created the YTD Sales measure using TOTALYTD with the sum of Amount and the Order Date field from List of Orders. The measure calculates year-to-date sales based on the date context available in the model.

10. Add the YTD Sales card

Observation: Added a Card visual using the YTD Sales measure and formatted its displayed value as currency. The card shows approximately ■158.8K in the captured report state and sits alongside the Average Profit in Delhi KPI.

11. Create the Sales Target Achievement by Category chart

Observation: Created a clustered column chart comparing Sum of Amount with Target by Category. Electronics, Clothing, and Furniture are displayed so the viewer can compare sales values against their respective target values.

12. Create the Max Profit Margin by Sub-Category donut chart

Observation: Created a donut chart with Sub-Category used for the legend and maximum Profit Margin used as the value. The chart presents a part-to-whole view of the sub-category values available in the data.

13. Format the Max Profit Margin chart

Observation: Adjusted the donut visual's title and layout and positioned it within the report. The legend identifies the sub-categories, while the segments and labels show the distribution of the selected maximum profit-margin values.

14. Create the Monthly Sales Trend line chart

Observation: Created a line chart using Month on the horizontal axis and Sum of Amount on the vertical axis. This visual makes month-to-month changes in sales easier to compare and helps identify higher and lower sales periods.

15. Create the Profit and Quantity Analysis scatter chart

Observation: Created a scatter chart to compare Quantity and Profit across the available product or sub-category groupings. The points help show how quantity and profit vary together and make potential patterns or outlying points easier to spot.

16. Arrange the main analytical charts

Observation: Resized and positioned the target comparison, profit-margin donut, monthly sales trend, and scatter chart. This layout step organized the main visuals into a dashboard structure so users can compare several aspects of sales performance in one view.

Charts and analytical visuals

17. Create the Sales Performance Matrix

Observation: Added a matrix visual with Category as rows, Sub-Category as columns, and Sum of Amount and Sum of Profit as values. The matrix provides a tabular comparison of sales and profit across product categories and sub-categories.

18. Position the Sales Performance Matrix

Observation: Placed the matrix alongside the other report visuals and adjusted its size and position. The matrix adds detailed values to complement the more visual chart types on the dashboard.

19. Create the Sales Distribution by Sub-Category treemap

Observation: Added a treemap to represent sales Amount across Category and Sub-Category. Treemap rectangles vary in size according to the value assigned, helping the viewer identify sub-categories that contribute larger or smaller portions of sales.

20. Complete and position the treemap

Observation: Configured the treemap to show the sub-category sales distribution and positioned it in the report layout. This gives the dashboard a compact view of the relative contribution of different sub-categories.

21. Create the Order Count Analysis visual

Observation: Created a funnel visual using Location as the category and the order-count field as the value. The visual compares the displayed order counts across locations, although the location field contains state/location labels rather than a separate Segment field.

22. Position the Order Count Analysis visual

Observation: Placed the order-count visual in the dashboard and checked that location labels and count values were visible. The visual complements the sales charts by showing how order counts vary across locations.

23. Configure the Sales by Location map

Observation: Added a map visual and configured it with the location field and sales Amount. The map is intended to show where sales occur geographically, using the available location values from the dataset.

Dashboard KPIs and final review

24. Display and position the Sales by Location map

Observation: Positioned the map with the other report visuals. The map shows geographic bubbles for locations represented in the dataset and provides a spatial view of sales alongside the monthly and category-based charts.

25. Add the Sum of Target card

Observation: Added a Card visual showing the sum of the Target column. In the captured report state, the card displays approximately 436K, giving a single headline value for the total target across the target records.

26. Create the Minimum Target measure

Observation: Created a Minimum Target measure using MIN on the Target column in the Sales target table. The measure returns the smallest target value in the current filter context.

27. Add the Minimum Target card

Observation: Added the Minimum Target measure to a Card visual. The displayed value is approximately ■9K in the captured report state, representing the overall minimum target from the available target records.

28. Format the Minimum Target card

Observation: Formatted and positioned the Minimum Target card to match the dark-blue theme and the other KPI cards. The card makes the minimum target value easy to identify in the dashboard.

29. Review the final dashboard

Observation: Reviewed the completed E-Commerce Sales Analytics report, including KPI cards, Category and Location slicers, sales-versus-target comparison, sales-distribution treemap, performance matrix, scatter chart, profit-margin donut, geographic map, monthly trend, and order-count visual. The available screenshots show no Segment field in the Sales target table, so the Minimum Target card represents the overall minimum target rather than minimum target by segment.

Final result: The report combines the calculated columns, DAX measures, KPI cards, interactive slicers, and sales-analysis visuals in one Power BI dashboard. Save the PBIX file and keep a backup before uploading the project and this observation document to GitHub.